

Data Handling: Import, Cleaning and Visualisation

Exercise/Workshop 3: Programming with Data

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Exercises

Exercise A: Write a Sum Function

In the lecture on Programming with Data, we have discussed the most basic programming concepts in R. Among other things, you have learned how we can use for-loops to sum up numbers stored in a numeric vector.

1. Use the code example in the respective lecture notes and the following empty function construct to implement a function that takes a numeric vector as input and returns the sum of the vector's elements.

```
my_sum <-  
  function(x){  
  
  }
```

2. Define the vector `numbers = c(1, 2, 3)`. Then, compare your function `my_sum()` with the `sum()` function (the pre-implemented sum function in R) when using `numbers` as the input argument: Do you get the same output for `my_sum(numbers)` and `sum(numbers)`?

Exercise B: Robustness and Warnings

1. Once you have implemented and successfully tested `my_sum()`, call your function with the following vector as input. What happens?

```
numbers2 <- c("1", "2", "3")
```

2. Why do we get an error? Investigate by comparing the vector `numbers` from Exercise A with the new vector `numbers2`. Hint: the functions `str()` and `class()` might help you.
3. Extend your `my_sum()` function with a control statement that checks whether the input is of the right class for the function to work (Hint: `class(x)!="numeric"`). The goal is to make the function more robust to wrong input. That is, if the input is wrong, issue a warning but don't break down with an error. Use the following code for the warning message:

```
warning("Wrong input! This function only accepts numeric or integer values!")
```

4. Test your enhanced sum function with the two input vectors `numbers` and `numbers2`.

Exercise C: Standard Deviation Function

1. During the lecture on Programming with Data we have step-by-step implemented our own R function to compute the mean. Now implement an R function called `my_sd` that computes the standard deviation,

defined as: $SD = \sqrt{\frac{1}{N-1} \sum_{i=1}^N (x_i - \bar{x})^2}$. Hint: revisit the math operators section in the notes on the first exercise session.

2. Test your implementation by comparing its output with the already implemented `sd()` function.

Additional Exercises (not exam-relevant)

Exercise D: Standard Error Function

Building on the function written in Exercise C, implement a function to compute (the estimate of) the sample mean standard error as defined here: $SE_{\bar{x}} = \frac{SD}{\sqrt{n}}$. Hint: the idea is to use your own implementation (in Exercise C) to compute SD as a first step of your standard error function.

Exercises C and D should illustrate how you can approach more complex problems by breaking them up in smaller problems and addressing each of these step-by-step. That is, when looking at the problem of computing $SE_{\bar{x}} = \frac{SD}{\sqrt{n}}$, you see that computing SD is a part of the overall problem, which can be addressed separately.

Exercise E: T-test

Implement a function for the one-sample t-test: $t = \frac{\bar{x} - \mu_0}{\frac{SE}{\sqrt{n}}}$.

Mock Exam Questions

Part 1: TRUE/FALSE Questions

For each of the following statements, decide whether the statement is TRUE (T) or FALSE (F).

1. 1GB = 1000 x 1000 x 1000 x 16 bits
-

Part 2: Multiple Choice Type A

For each of the following questions/statements, decide which of the possible responses are correct. NOTE: Either ONE OR SEVERAL of the response statements can be correct. In order to get a point, all correct response statements need to be marked!

1. In csv files, rows of data are indicated by new lines. Therefore...

- A) A special character marks the end of a line. This is why a CSV-parser recognizes when a line ends.
 - B) CSV files can only be opened in spreadsheet software like Excel.
 - C) The underlying binary code contains white spaces whenever an end of line occurs
 - D) Computers recognize the end of the line because each line contains the same amount of characters
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Part 3: Multiple Choice Type B (only one correct!)

For each of the following questions/statements, decide which of the possible responses is correct. NOTE: For each question only ONE of the possible responses is correct. Select ONLY ONE of the response statements per question!

1. Look at the following line of code. Which of the four statements about the code is correct?

```
<html><head><title>alpha</title></head><head><body>beta</body></head></html>
```

- A) The <title> tag should be placed in between the <body> tags in order to make this code a valid HTML document
- B) The code represents a basically correctly specified HTML-document
- C) All tags occurring in the code are valid HTML tags
- D) 'alpha' and 'beta' are HTML-tag attributes